Sequences & Series

Section 1: Advanced Arithmetic & Geometric Progressions

1 The sum of three consecutive terms of an arithmetic progression is 27, and their product is 504. What is the largest of the three terms?

a. 14

b. 16

c. 18

d. 21

2 If x, y, z are in geometric progression, which equation must be true?

a. $y^2 = xz$

b. $2y = x + z$

c. $y = \frac{x+z}{2}$

d. $y^3 = xz$

3 What is the sum of the series $1 + \frac{2}{3} + \frac{4}{9} + \frac{8}{27} + \dots$?

a. 3

b. $\frac{3}{2}$

c. 2

d. $\frac{5}{2}$

Section 2: Summation Notation & Formulas

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4 Evaluate $\sum_{k=1}^{15} (4k - 1)$.

a. 465

b. 480

c. 495

d. 510

5 Evaluate $\sum_{k=1}^{8} k^3$ using the formula $[\frac{n(n+1)}{2}]^2$.

a. $1,296$

b. $1,024$

c. $1,521$

d. $1,728$

6 Evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

a. 1

b. $\frac{1}{2}$

c. 2

d. Diverges

Section 3: Infinite Geometric Series & Repeating Decimals

7 A ball is dropped from a height of 16 meters. Each time it hits the ground, it bounces to $\frac{3}{4}$ of its previous height. What is the total vertical distance traveled before coming to rest?

a. 112 meters

b. 96 meters

c. 128 meters

d. 64 meters

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8 Express $0.\overline{36}$ as a reduced fraction.

a. $\frac{4}{11}$

b. $\frac{36}{100}$

c. $\frac{1}{3}$

d. $\frac{12}{33}$

9 For what range of x does $\sum_{n=0}^{\infty} (\frac{x}{3})^n$ converge?

a. $(-3, 3)$

b. $[-3, 3]$

c. $(-1, 1)$

d. $(-9, 9)$

Section 4: Mathematical Induction

10 In proving $1 + 2 + \ldots + n = \frac{n(n+1)}{2}$, what is the inductive step?

a. Add $(k + 1)$ to both sides of the hypothesis for $n = k$

b. Multiply both sides by 2

c. Subtract k from n

d. Differentiate with respect to n

11 What is the base case when proving a theorem for all positive integers n?

a. Verify the statement for $n = 1$

b. Verify for $n = 0$

c. Assume for $n = k$

d. Prove for all even integers

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12 If $P(k) \Rightarrow P(k+1)$ is true, under what condition is $P(n)$ true for all $n \geq 1$?

a. $P(1)$ is true

b. $P(0)$ is true

c. $P(2)$ is false

d. $P(k)$ is true for all k simultaneously

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